

Chapter 1

1) Architecture of .NET Framework (5 Marks)

- The .NET Framework architecture consists of the **CLR (Common Language Runtime)**, which manages execution of .NET programs.
 - **Base Class Library (BCL)** provides pre-built classes for I/O, data, networking, security, etc.
 - **Languages** like C#, VB.NET, and F# can be used because of the Common Language Specification (CLS).
 - **ASP.NET** is part of the framework for web application development.
 - **ADO.NET** supports data access for connected and disconnected environments.
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2) Explain Master Page (4 Marks)

- Master Page allows **consistent layout** across multiple pages in a web application.
 - It contains **placeholders (ContentPlaceHolder)** for child pages.
 - Child pages use **Content controls** to define their unique content.
 - Example: A website with a common header/footer in all pages.
 - Helps in **code reusability and maintenance**.
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3) What is CSS? Types of CSS and CSS in Themes (4 Marks)

- CSS (Cascading Style Sheets) is used to **design and format HTML elements**.
 - It separates **content from presentation**.
 - Types of CSS:
 1. Inline CSS – written inside an element's style attribute.
 2. Internal CSS – defined inside <style> in the <head> section.
 3. External CSS – written in .css file and linked with <link>.
 - **CSS in Themes:** ASP.NET themes allow applying consistent CSS, skins, and appearance across a website.
 - Themes are stored in the **App_Themes** folder.
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Chapter 2

1) Explain DataAdapter, DataReader, DataSet

- **DataAdapter:** Acts as a bridge between database and DataSet. Supports disconnected data.
 - **DataReader:** Provides fast, forward-only, read-only access to data. Best for performance.
 - **DataSet:** In-memory representation of data, can store multiple tables.
 - DataAdapter uses commands (Select, Insert, Update, Delete) to update DataSet.
 - DataReader is best for **large amount of data reading**.
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2) Database Access using ADO.NET (5 Marks)

- ADO.NET is a **data access technology** in .NET.
 - Provides connected (**DataReader**) and disconnected (**DataAdapter + DataSet**) models.
 - Steps to access database:
 1. Create **SqlConnection** with connection string.
 2. Use **SqlCommand** to execute queries.
 3. Fetch data using **SqlDataReader** or **DataAdapter**.
 4. Fill data into **DataSet/DataTable**.
 5. Close connection to release resources.
 - Supports multiple databases like SQL Server, Oracle, etc.
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3) Explain Cookies and QueryString with Example

- **Cookies:**
 - Store small pieces of data in client's browser.
 - Can be **persistent (saved)** or **non-persistent (session-based)**.
 - Example: `Response.Cookies["user"].Value = "John";`
 - Useful for login tracking, preferences, etc.
- **QueryString:**
 - Pass data between pages via URL.
 - Example: `page2.aspx?user=John`
 - Retrieved using: `Request.QueryString["user"]`

- Simple but less secure (visible in URL).
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Chapter 3

1) Write a short note on Web.config file with Example (4 Marks)

- **Web.config** is an XML file for ASP.NET application configuration.
 - Stores settings like connection strings, authentication, authorization, custom errors.
 - Example:
 - <configuration>
 - <connectionStrings>
 - <add name="con" connectionString="Data Source=.;Initial Catalog=MyDB;Integrated Security=True"/>
 - </connectionStrings>
 - </configuration>
 - Provides **centralized configuration** for the application.
 - Each application can have its own Web.config file.
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2) Short Note on Web Services (IMP)

- Web Services are software components accessible over the internet.
- They use **XML-based communication (SOAP)** or **REST (JSON/HTTP)**.
- Platform independent – developed in one language, used in another.
- Example: Currency converter, weather report service.
- ASP.NET supports **ASMX Web Services** and **WCF (Windows Communication Foundation)**.

3) Explain Session and Cookies in ASP.NET (5 Marks)

- Cookies are small text files stored on the client browser to maintain user-specific information.
- They can be temporary (session cookies) or permanent (persistent cookies with expiry date).
- Cookies are lightweight but less secure, as data is stored on the client side.
- Example for cookies: `Response.Cookies["UserName"].Value = "John";`

- Session is a server-side state management technique to store user-specific data.
- Each user is assigned a unique SessionID (stored in a cookie or URL).
- Sessions are more secure than cookies because data is stored on the server.
- Default session timeout is 20 minutes, configurable in Web.config file.
- Example for session: `Session["UserName"] = "John";`